

RGV Water GIS Command Center - User Manual

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RGV Water GIS Command Center — User Manual

This manual is for planners, analysts, and stakeholders using the RGV Water GIS Command Center day-to-day.

1) Overview

What this system does

- ? Provides a secure GIS map workspace for the Rio Grande Valley (RGV).
- ? Lets you create/edit infrastructure features (points + lines) and assign them to layers and projects.
- ? Adds planning tools (route alternatives, cost estimation, ROW corridors, overlays, imported reference layers).
- ? Adds decision dashboards for executive/investor/legislative communication.

Data ownership / privacy

- ? Data is user-scoped by default (you see what you create under your login).
- ? If you log in with a different account, you will not see your prior account's records.

2) Getting Started

2.1 Sign in

1. Open the app URL provided by your administrator.
2. Go to the Login screen.
3. Sign up or sign in using your email/password.

If you can't sign in

- ? Confirm with your admin that Email auth is enabled in Supabase Auth.
- ? If email confirmation is required, verify your email first.

Screenshot placeholder: Login screen (email + password)

2.2 Navigation (Where to go)

- ? Map: the primary GIS workspace (dashboard/map view).
- ? Planning Tools: Imported Layers, Map Overlays, File Uploads, ROW Corridors, Route Comparison, Cost Estimator, Exports.
- ? Dashboards: Executive Summary, Investor Snapshot, Project Performance, Legislative Briefs, Dashboard Exports.

Screenshot placeholder: Left navigation menu

3) Dashboard Overview

3.1 Executive Summary

Purpose

? Portfolio totals across projects, cost estimates, CRM follow-ups, and upcoming funding deadlines.

How to use

1. Open Dashboards !' Executive Summary.
2. Review key totals: Projects, Demand (MGD), Revenue, Capex (Estimated), Follow-ups, Funding deadlines.
3. Use Refresh to reload.

Screenshot placeholder: Executive Summary tiles + Upcoming Funding Deadlines

3.2 Investor Snapshot

Purpose

? Ranks projects by revenue and shows ROI as $\frac{\text{Revenue}}{\text{Capex}}$ using the latest cost estimate per project.

How to use

1. Open Dashboards !' Investor Snapshot.
2. Use Refresh to reload.
3. Review the table for Revenue, Capex, ROI.

Screenshot placeholder: Investor Snapshot table

3.3 Project Performance

Purpose

? Quick health view: counts by status and priority, plus recently updated projects.

How to use

1. Open Dashboards !' Project Performance.
2. Use Refresh to reload.
3. Review the "By Status" and "By Priority" cards.

Screenshot placeholder: Project Performance (By Status / By Priority)

4) Maps

4.1 Layer Controls

The Map sidebar begins with Layer Controls.

Choose a layer (what you are adding/editing)

1. Use the Choose a layer dropdown.
2. The selected layer is the target for new/edited features.

Show/hide layers

? In the layer list, toggle the checkbox next to a layer to show/hide it.

Add a layer

1. Click Add Layer.
2. Enter New layer name.
3. Click Save.

Delete a layer

? Click Delete Layer (for the active layer) or click the × next to a layer in the list.

? You must keep at least one layer.

Screenshot placeholder: Layer Controls (dropdown + Add Layer + layer list)

4.2 Add and edit map features (Points and Lines)

Create a point

1. Click Point.
2. Click once on the map to place the point.
3. Fill in Feature Details (title, notes, project, etc.).
4. Click Save.

Create a line (pipeline route)

1. Click Line.
2. Click to add vertices along the route.
3. Double-click the last point to finish.
4. Fill in Feature Details.
5. Click Save.

Move/edit geometry

1. Click an item on the map (or from Items In Selected Layer).
2. Drag to adjust geometry.
3. Click Save.

Delete an item

1. Click the × next to the item in Items In Selected Layer.
2. Confirm removal.
3. Click Save to sync deletions to the database.

Screenshot placeholder: Map with a selected feature + Feature Details panel

4.3 Feature Details (metadata)

When a feature is selected, you can set:

- ? Layer (move the feature to a different layer)
- ? Project (associate with a project)
- ? Feature title and Notes
- ? Estimated construction cost (USD)
- ? Priority (e.g., HIGH/MEDIUM/LOW)
- ? Contact name / Phone / Email
- ? Source URL

Route distance

- ? When a line feature is selected, the sidebar shows Route distance: X miles.

Screenshot placeholder: Feature Details fields list

4.4 Map exports

GeoJSON export (current layer set)

1. Click GeoJSON.

2. A file named rgv-water-gis-export.geojson downloads.

Screenshot placeholder: Export buttons (Save / GeoJSON)

5) Layers (User-Managed)

5.1 When to create a new layer

Use layers to separate different planning “themes,” such as:

- ? Desal plants
- ? Pipelines / transmission
- ? Storage tanks
- ? Candidate wellfields
- ? Constraints / issues

5.2 Recommended layer naming

- ? Keep labels short and scannable.
- ? Use a consistent naming scheme across teams (e.g., Pipelines — South, Pipelines — North).

6) Stakeholders (CRM)

6.1 Stakeholder CRM (Organizations)

Purpose

- ? Track utilities, cities, counties, agencies, and partners as Organizations.

How to use

1. Open Stakeholder CRM.
2. Enter New organization name.
3. Enter Type (e.g., utility, city, county).
4. Click Add.
5. To remove, click Delete.

Screenshot placeholder: Stakeholder CRM list + Add form

Note on contacts/meetings/notes

- ? The database contains tables for contacts, meetings, and notes; the current UI focuses on Organizations (the foundation record).

7) Funding Database

7.1 Funding programs

Purpose

- ? Track program name, agency, deadline, and URL.

How to use

1. Open Funding.
2. Fill in:
 - Program name

- Agency
- Deadline (YYYY-MM-DD)
- URL
- 3. Click Add Program.
- 4. To remove a program, click Delete.

Screenshot placeholder: Funding programs list

How it connects to dashboards

? Executive Summary shows upcoming deadlines in the next 30 days.

8) Pipeline Route Builder

There are two route-related workflows:

8.1 Quick route sketching on the Map

Purpose

? Draw a line on the Map as a pipeline concept and store it as a GIS feature.

Steps

1. Open Map.
2. Select an appropriate layer.
3. Click Line.
4. Draw the route and double-click to finish.
5. In Feature Details, choose a Project (optional but recommended).
6. Click Save.

Tip: Select the line later to view distance in miles.

8.2 Route Comparison tool (A/B/C alternatives)

Purpose

? Create named alternatives (Route A/B/C) for a specific project and compare length and cost.

Steps

1. Open Planning Tools !' Route Comparison.
 2. Select a Project.
 3. Click New Alternative.
 4. Draw the route line on the map.
 5. Fill in inputs (as available):
- Name (A/B/C or label)
 - Cost per mile
 - Crossings
 - Easement concerns
 - Permitting concerns
 - Environmental concerns
6. Click Save.

Screenshot placeholder: Route Comparison (project selector + map)

9) Cost Estimator

9.1 Create or update a cost estimate

Purpose

? Estimate a total cost using pipeline miles, cost-per-mile, lump sums, and percentage adders.

Steps

1. Open Planning Tools !' Cost Estimator.
2. Select a Project.
3. Enter values:
 - Pipeline miles
 - Cost per mile
 - Pump station cost
 - Storage tank cost
 - Land / easement cost
 - Engineering/design % (optional)
 - Permitting/environmental % (optional)
 - Contingency % (optional)
4. Review Totals (Base + Total).
5. Click Save Estimate.

Example workflow

- ? Use Route Comparison length (miles) to seed Pipeline miles.
- ? Use engineering/permitting/contingency percentages for concept-level ranges.

Screenshot placeholder: Cost Estimator (inputs + Totals)

10) Scenario Builder (Phase 4 — Database Ready)

Status

- ? Phase 4 installs the Scenario tables in the database.
- ? A dedicated Scenario Builder UI is planned (Phase 4.x).

What scenarios will do (intended)

- ? Compare “bundles” of utilities/routes/capex under a scenario name.

Current workaround (until Scenario UI ships)

- ? Use Projects + Route Comparison + Cost Estimator as the working scenario proxy.
- ? Use file uploads to attach supporting assumptions.

11) Revenue Model (Phase 4 — Database Ready)

Status

- ? Phase 4 installs revenue_models for storing assumptions.
- ? A Revenue Model UI is planned (Phase 4.x).

Current workaround

- ? Populate Project revenue for dashboard ROI.

? Attach spreadsheets and assumptions under File Uploads or Documents.

12) Risk Register (Phase 4 — Database Ready)

Status

- ? Phase 4 installs risk_register.
- ? A Risk Register UI is planned (Phase 4.x).

Current workaround

- ? Maintain risks as uploaded documents/spreadsheets.
- ? Summarize key risks in Legislative Briefs until the UI is available.

13) Meeting Brief Generator

13.1 Legislative Briefs

Purpose

- ? Generate and store a policy-ready draft brief for a project.

Steps

1. Open Dashboards !' Legislative Briefs.
2. Choose a Project.
3. Optionally choose an Organization.
4. Optionally enter a Title.
5. Click Generate Draft.
6. Edit Brief Text as needed.
7. Click Save Brief.
8. To re-open a saved brief, click an entry under Recent Saved Briefs.

Screenshot placeholder: Legislative Brief editor + saved list

14) Document Repository

14.1 Project documents

Purpose

- ? Store per-project documents in Supabase Storage under documents/project/<project_id>/....

Steps

1. Open Documents.
2. Choose a Project.
3. Under Upload, select a file.
4. Wait for "Uploaded."
5. To delete, click Delete next to a document.

Screenshot placeholder: Documents (project picker + file list)

15) File Uploads (Planning Uploads)

15.1 Upload planning files

Purpose

? Upload GIS and supporting files and tag them to project/organization/county/district/utility.

Supported file types

? PDF, PNG, JPG/JPEG

? GeoJSON/JSON

? KML/KMZ

? Shapefile ZIP

Steps

1. Open Planning Tools !' File Uploads.
2. (Optional) Choose Project and/or Organization.
3. (Optional) Fill in County, District, Utility, Notes.
4. Select a File.
5. Click Upload.

Delete an upload

? Click Delete next to a file.

Screenshot placeholder: File Uploads form

16) Imported Layers (Reference GIS)

16.1 Import GIS files into read-only map overlays

Purpose

? Import external GIS layers into PostGIS for reference (not editable from the Map).

Supported import types

? GeoJSON, KML, KMZ, zipped shapefile (.zip)

Steps

1. Open Planning Tools !' Imported Layers.
2. Enter Layer name (example: "FEMA Floodplain").
3. Choose either:
 - Import from existing upload, or
 - Or upload a new GIS file
4. Click Import Layer.
5. Return to the Map sidebar to toggle the imported layer on/off.

Performance guardrail

? Imports are limited to 5000 features per layer for performance.

Screenshot placeholder: Imported Layers import form

17) Map Overlays (PDF / Image Overlays)

17.1 Create a raster overlay

Purpose

? Place a PDF (first page) or image as a semi-transparent overlay on the map, then georeference it using corner coordinates.

Steps (high level)

1. Open Planning Tools !' Map Overlays.
 2. Create an overlay from an existing upload or a new file.
- PDFs are automatically converted to a PNG of the first page.
3. Select the overlay and set its corners by clicking four points on the map.
 4. Adjust opacity as needed.
 5. Return to the Map sidebar to toggle overlay visibility.

Screenshot placeholder: Map Overlays creation + corner placement

18) ROW Corridors

18.1 Create or edit a ROW corridor

Purpose

? Store right-of-way corridors (point/line/polygon) with metadata and map visibility toggles.

Steps

1. Open Planning Tools !' ROW Corridors.
2. Draw a corridor geometry (point/line/polygon) using the draw tools.
3. Fill in:
 - Name
 - Corridor type
 - Corridor owner (optional)
 - Width (ft) (optional)
 - Source (optional)
 - Verification status (unverified/partial/verified)
 - Notes (optional)
4. Click Save.

Visibility

? Use the corridor list toggles to show/hide corridors on the map.

Screenshot placeholder: ROW Corridors editor

19) Export Center

19.1 Dashboard exports

Purpose

? Download sharable data files for leadership/investor workflows.

Exports available

- ? Executive Summary JSON
- ? Investor Snapshot CSV

Steps

1. Open Dashboards !' Dashboard Exports.
2. Click Download JSON or Download CSV.
3. If export persistence is enabled, the export is also uploaded to Supabase Storage (planning_exports).

Screenshot placeholder: Dashboard Exports cards

19.2 Planning exports (coming online)

? Planning Tools !' Exports currently describes the intended export set export launcher + history screen.

20) Troubleshooting

20.1 “Load failed... Did you run supabase/phaseX.sql?”

- ? This usually means the database tables/RPC/policies for that module are not installed yet.
- ? Notify your admin to run the appropriate SQL migration (Phase 2/3/4).

20.2 I can't see data my teammate created

- ? Data is user-scoped. Confirm you are using the same account or your admin has implemented shared/role-based access.

20.3 GeoJSON/KML import fails

- ? Confirm the file contains features.
- ? Split large files to stay under 5000 features.

20.4 Overlay says “(set corners)”

- ? The overlay exists, but its georeferencing corners are not yet set.
- ? Go to Map Overlays and place the four corners.

21) FAQ

Q: What coordinate system is used?

- ? GIS features are stored in WGS84 (EPSG:4326).

Q: Can I export everything to share externally?

- ? Map GeoJSON export is available now.
- ? Dashboard exports (JSON/CSV) are available now.
- ? Full planning export packs (PDFs, map renders) are planned.

Q: Are imported layers editable?

- ? No. Imported Layers are read-only reference overlays.